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Development and validation of a deep learning-based emergency triage model: a feasibility and effectiveness study

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Abstract

Background Due to insufficient triage accuracy, general purpose Large Language Models (LLMs) cannot be directly applied to clinical pre-triage. At the same time, the existing emergency pre-triage systems also have certain limitations. Dedicated self-trained models, with their advantages of high performance, easy scalability, and low cost, have become the primary development direction in the field of medical diagnosis. We aimed to propose a method for training a dedicated emergency triage model to improve the accuracy of triage level classification for emergency patients.

Methods This study employed emergency triage data from December 2023 to December 2024 (raw data: 270,298 entries, cleaned data: 113,302 entries). The data was temporally partitioned into a training set ($n = 91{,}743$; 8:2 split for training and validation), a test set ($n = 11{,}559$), and an online diagnostic set ($n = 10{,}000$). The data was uploaded to the EasyDL platform to develop and deploy an emergency triage model, followed by the evaluation of model performance and online diagnosis.

Results The emergency triage model developed on the EasyDL platform demonstrated reliable performance within defined parameters. The model achieved 97.2% accuracy (95% CI: 96.2–98.2%) on the training set and 97.7% (95% CI: 96.8–98.6%) on the test set. Post-deployment, the overall triage accuracy reached 89.45%, with > 90% accuracy across all disease severity levels (I–IV). The Matthews Correlation Coefficient (MCC) was 0.92. External validation achieved an accuracy of 78.3% (95% CI: 76.9–79.8%), representing a modest decrease compared to the Online Diagnostic set. The development process was simple and efficient; the model can be deployed to mobile terminals via an Application Programming Interface (API), enabling direct clinical application.

Conclusion The deep learning-based emergency triage model demonstrates potential for clinical translation, and its development plan is operationally feasible. However, to realize its practical value, subsequent research and efficacy validation in real-world clinical settings are necessary prior to implementation.

Keywords Large language models, Emergency medicine, Workflow efficiency, Artificial intelligence, AI

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Introduction

Triage serves as the critical first point of contact in healthcare delivery, where its accuracy and efficiency directly influence patient outcomes and the rational allocation of medical resources. Conventional experience-based triage models are increasingly inadequate in addressing the demands of modern emergency care, particularly in terms of timeliness and the management of fragmented patient information. Furthermore, existing rule-based structured triage systems [1] exhibit several limitations, including difficulties in data standardization, limited regional adaptability, high operational costs, and inherent inflexibility [2]. These shortcomings contribute to suboptimal assessment performance for time-sensitive and complex conditions, including acute chest pain [3]. Therefore, the exploration of innovative technologies to enhance triage performance has become a pressing research priority in emergency medicine.

Large language models (LLMs) have demonstrated substantial potential in optimizing clinical workflows and enhancing decision-making quality [4, 5]. Most existing studies have employed text-based inputs and outputs to evaluate the diagnostic performance of LLMs. In three ophthalmology studies [6–8], two general internal medicine studies [9, 10], and one each in radiology [11] and general practice [12], the GPT series of LLMs achieved diagnostic accuracies exceeding 80%. In the domain of emergency triage, a prospective study by Paslı et al. [13] demonstrated that the general GPT-4 model exhibited a high level of agreement with professional medical teams in triage decisions (Kappa = 0.899), indicating the viability of LLMs for this application. Furthermore, Williams et al. [14] validated the model's capability to assess relative disease severity through paired comparison tasks, achieving an accuracy of 89%. Nevertheless, the performance on such simplified tasks does not fully represent the model's competency in the complex, multi-level triage decision-making required in clinical practice. In a separate effort, Yazaki et al. [15] aimed to enhance the triage performance of general-purpose LLMs using retrieval-augmented generation (RAG) technology, reporting an improved accuracy of 70%. While this outcome marks a notable technological advancement, the achieved performance still falls short of the requisite standards for high-precision clinical decision support. This disparity between theoretical potential and practical effectiveness stems largely from the general-purpose model's lack of domain-specific training in emergency triage protocols and its insufficient integration into established clinical workflows [16]. Consequently, current models fall short of the rigorous demands of clinical decision-making regarding generalizability, interpretability, and risk controllability. The successful clinical translation of these systems necessitates the development of triage-specific models—fine-tuned on local data, embedded with clinical reasoning, and equipped with human-AI collaboration mechanisms—rather than the direct deployment of general-purpose LLMs.

This study introduces a specialized training methodology for an emergency pre-triage model. By leveraging standardized triage data from hospital information systems, this approach algorithmically transforms static data [17, 18] into dynamic knowledge [19, 20], thereby exploring a viable path toward improving triage accuracy for emergency patients. Considering that clinical practitioners generally lack specialized training in algorithm development, this study employed an accessible artificial intelligence development platform. This strategy maintained model performance while minimizing technical complexities in training, deployment, and clinical implementation, thereby enhancing the method's accessibility and practical feasibility in real-world healthcare environments.

Methods

Study design and data source

This study utilized a convenience sampling approach. Data was retrieved from the hospital's electronic triage decision-support system, resulting in an Excel dataset containing emergency patient triage records collected from December 2023 to December 2024. The final data set consisted of 270,298 triage entries across 43 data fields.

Inclusion criteria were: (1) patients registered for emergency appointment-based medical services; and (2) age ≥ 18 years. **Exclusion criteria** included: (1) appointment cancellations; (2) missing critical data (e.g., patient ID, method of arrival, system classification, triage result etc.); and (3) withdrawal or refusal (e.g., refusal of vital sign measurement, request for prescription or diagnostic document, or referral to external facilities etc.). In accordance with the four-level emergency triage standard [21] and the Expert Consensus on Emergency Triage [22], the raw data underwent preprocessing, including removal of sensitive information, data consolidation, and standardization of formatting. Fourteen key variables were retained: sex, age, consciousness, body temperature, pulse, respiratory rate, blood pressure (2 items, Systolic blood pressure and Diastolic blood pressure, SBP & DBP), Saturation of peripheral oxygen (SpO_2), pain score, method of arrival, and chief complaint (3 items, classification–chief complaint–basis for Judgement; e.g., Respiratory system–shortness of breath–moderate respiratory distress). After processing, a total of 113,302 records were included. The data filtering process is illustrated in Fig. 1.

The dataset was chronologically divided into Training Set, Test Set, and Online Diagnostic Dataset. From

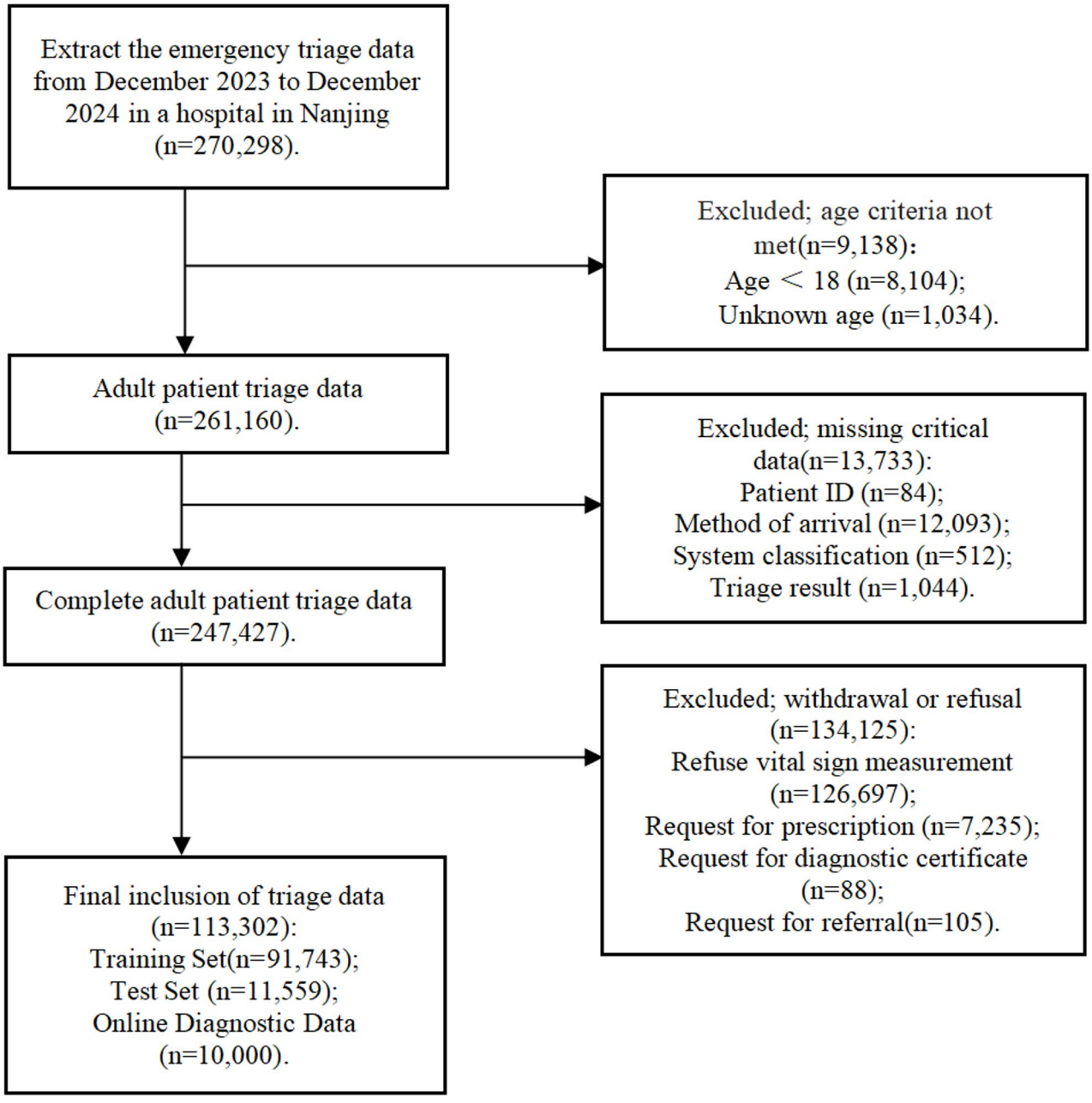

Fig. 1 Flow diagram

December 2023 to October 2024, a total of 91,743 records were split by the platform into training and validation sets at an 80:20 ratio. A separate set of 11,559 triage records from November 2024 was designated as the test set to objectively assess model performance on unseen data. Furthermore, 10,000 records from December 2024 served as post-deployment data to evaluate the model's real-world classification performance. To externally validate the generalizability of this model, 4,504 triage records were collected from another tertiary hospital in the same region between December 11–18, 2025.

Emergency triage criteria

The four-level emergency triage standard implemented at Nanjing Drum Tower Hospital is grounded in the scientific framework and assessment logic of the Taiwan Triage and Acuity Scale (TTAS) [23] and adheres to relevant national guidelines, including the 2011 *Guiding Principles for Emergency Patient Triage Classification (Draft for Comments)* issued by the National Health Commission of China [24]. To address local clinical needs, the original five-level TTAS structure was consolidated and refined through a literature review and a Delphi process, resulting in a widely applicable four-level (I–IV) triage

management standard [25]. Since being implemented clinically in 2013 as an electronic decision-support system, this standard has been used to triage patients based on five vital sign measurements, the Glasgow Coma Scale (GCS), trauma scores, and patient-reported chief complaints for rapid, accurate acuity evaluation.

Stability of the triage protocol

The formal triage protocol, including its classification criteria and scoring algorithms, remained unchanged at the institutional level throughout the study period and was consistently applied by senior triage nurses. Consequently, any temporal variations observed in the patient population (e.g., in demographics or case mix) are interpreted as natural clinical or seasonal fluctuations, rather than artifacts of changes in the triage protocol itself. The triage level documented in the electronic information system served as the reference standard for evaluating the model's predictions. This consistency in the reference labels ensures that model training and evaluation were conducted within a fixed and reliable framework.

Data preprocessing and feature engineering

To leverage the robust semantic representation capabilities of pre-trained ERNIE model (Enhanced Representation through kNowledge IntEgration, Baidu Inc.) within the EasyDL platform's text-classification framework, structured clinical variables (including categorical and continuous types) were transformed into a unified textual format. This serialization approach allows the model to interpret the contextual relationships between clinical features—such as the association between age, vital signs, and chief complaint—in a manner analogous to natural clinical narrative, rather than treating them as independent numeric inputs. Accordingly, a customized prompt protocol was developed using the GPT-4 API (OpenAI; February–March 2025) to systematically extract and serialize entities from Microsoft Excel sources into semicolon-delimited text strings (e.g., Gender: Male; Age: 63 years; Temperature: 36.9 °C; Pulse: 105 beats/min). The concatenated text was tokenized using the standard ERNIE tokenizer (Baidu Inc.) subword segmentation and maps tokens to vocabulary IDs for embedding lookup, thereby aligning the input with the model's expected format. The reliability of GPT-4's outputs was ensured through independent cross-validation by two research assistants, verified by a 20% random audit confirming an error rate < 0.01%.

Model construction and deployment

Model architecture, hyperparameters, and training configuration

This study was conducted using the short-text classification framework on the EasyDL platform. The underlying network was ERNIE 2.0 Base (Chinese), an architecture isomorphic to BERT-Base that features a 12-layer bidirectional Transformer encoder (hidden size = 768, attention heads = 12, and ~110 million parameters). Model selection was limited by the computational resources and pre-trained model availability of our research platform. Training was performed on a single Tesla V100 Graphics Processing Unit(GPU) (Tesla V100, 16 GB). As the platform's default hyperparameters have been established on large-scale Chinese tasks, no manual grid search was performed. The detailed training configuration is summarized below:

Optimizer AdamW

Initial learning rate 3×10^{-5} (with linear decay)

Global batch size 32

Maximum sequence length 25

Training epochs 3

Early stopping patience 1

Warm-up ratio 0.1

Weight decay 0.01

Dropout rate 0.1

Training preparation phase

A total of 91,743 records were designated as the emergency triage Training Set, comprising 684 Level I: Critical (0.7%), 7,965 Level II: Severe (8.7%), 34,090 Level III: Urgent (37.2%), and 49,004 Level IV: Less-Urgent (53.4%) cases. An additional 11,559 records were designated as the test set, containing 83 Level I (0.7%), 766 Level II (6.6%), 4364 Level III (37.8%), and 6346 Level IV (54.9%) cases. Due to the imbalance in the number of samples across different triage levels and to avoid mistakenly deleting similar but non-duplicate samples, the uploaded data was processed without deduplication. A unified annotation system was established, comprising four triage acuity levels: Level I (Critical), Level II (Severe), Level III (Urgent), and Level IV (Less-urgent). The training and test sets were uploaded to the platform and manually annotated by two senior nurses, each with over five years of emergency triage experience. Annotations were performed with reference to ground-truth triage results obtained through a structured data management process.

Model training and validation phase

Relevant model information was provided, including name, industry type, application scenario, and business description. Training was conducted using the private dataset and deployed to the public cloud. A high-accuracy algorithm was selected, with Precision and Recall as the primary evaluation metrics. The training environment was conducted on a V100 GPU, which delivered a computational performance of 14 TeraFLOPS (Floating-point Operations Per Second), achieving a speed-up ratio of 1.5 over the baseline CPU.

Model deployment phase

The model inference endpoint was configured at https://aip.baidubce.com/rpc/2.0/ai_custom/v1/text_cls/EmergencyS11.The trained model was deployed to the public cloud API server via the EasyDL platform. An application was created to obtain the API Key and Access Code, which serve as the unique credentials for accessing the service.

Online application testing

API was accessed via HTTP POST requests, with the service name and endpoint suffix (EmergencyS11) properly configured. Two researchers manually input 10,000 records from the online diagnostic dataset to simulate real-world usage after model deployment. The model processed each entry individually, with per-record inference time ranging from 289 to 660 milliseconds.The online diagnostic results in Fig. 2.

External validation

A total of 4504 records were designated as the external validation set, comprising 27 Level I: Critical (0.6%), 410 Level II: Severe (9.1%), 1,752 Level III: Urgent (38.9%) and 2,315 Level IV: Less-Urgent (51.4%) cases. All records were tested individually following model deployment.

Baseline models and configuration

To benchmark the performance of our deep learning model, we implemented three widely-used machine learning algorithms: Logistic Regression, Random Forest, and Extreme Gradient Boosting (XGBoost).


```
{
  "text": "性别：男；年龄：94岁；体温：36.5°C；脉搏：85次/min；呼吸：N/A次/min；血压：117/82mmHg；SpO2：99%；NRS评分：0分；入院方式：120；预检：神经系统-步态失调/运动失调-意识程度改变（GCS 9-13）。"
}
{
  "log_id": "1087709315317115627",
  "results": [
    {
      "name": "二级急重",
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    },
    {
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    },
    {
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      "score": "0.0003441844310145825"
    },
    {
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      "score": "0.00009783235873328522"
    }
  ]
}
```
Level II: Severe 0.998
Level III: Urgent 0.001
Level IV: Less-Urgent 0.000
Level I: Critical 0.000

```
{
  "text": "286. 性别：女；年龄：73岁；体温：N/A°C；脉搏：78次/min；呼吸：18次/min；血压：106/57mmHg；SpO2：62%；NRS评分：0分；入院方式：120；预检：呼吸系统-呼吸短促-有气喘且FEV1或PEFR<40%的预期值（重度）。"
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    {
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    },
    {
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    },
    {
      "name": "四级亚急症",
      "score": "0.00010964206012431532"
    }
  ]
}
```
Level I: Critical 0.996
Level II: Severe 0.003
Level III: Urgent 0.000
Level IV: Less-Urgent 0.000

```
{
  "text": "性别：男；年龄：80岁；体温：N/A°C；脉搏：92次/min；呼吸：15次/min；血压：182/97mmHg；SpO2：96%；NRS评分：0分；入院方式：120；预检：心脏血管系统-心因性胸痛/胸闷-心因性疼痛已缓解/有典型心因性病史。"
}
{
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  "results": [
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    },
    {
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    },
    {
      "name": "四级亚急症",
      "score": "0.0008475962677039206"
    },
    {
      "name": "一级急危",
      "score": "0.00007676758832531 05"
    }
  ]
}
```
Level III: Urgent 0.976
Level II: Severe 0.022
Level IV: Less-Urgent 0.000
Level I: Critical 0.000

```
{
  "text": "性别：女；年龄：87岁；体温：N/A°C；脉搏：80次/min；呼吸：17次/min；血压：135/85mmHg；SpO2：98%；NRS评分：0分；入院方式：120；预检：肠胃系统-腹痛-急性中枢轻度疼痛（<4）。"
}
{
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    },
    {
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    },
    {
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      "score": "0.00009772175690159202"
    },
    {
      "name": "一级急危",
      "score": "0.000027741349185816944"
    }
  ]
}
```
Level IV: Less-Urgent 0.999
Level III: Urgent 0.000
Level II: Severe 0.000
Level I: Critical 0.000

Fig. 2 The online diagnostic results

For the Logistic Regression (logit) model, we utilized the L-BFGS optimizer with a maximum iteration limit of 1000 (max_iter = 1000). To address class imbalance, we set class_weight='balanced', and the regularization strength was assigned the default value (C = 1.0).

For the Random Forest model, we employed fixed hyperparameters without grid search: the number of trees was set to n_estimators = 100, the maximum tree depth was unlimited (max_depth = None), and class_weight='balanced' was applied to address class imbalance. The random seed was fixed (random_state = 42) for reproducibility.

For XGBoost model, we employed a fixed configuration without hyperparameter search: learning_rate = 0.1, max_depth = 6, and n_estimators = 200. Additional parameters were set as subsample = 0.8, colsample_bytree = 0.8, and reg_alpha = 0.1. These values were selected based on empirical considerations rather than systematic grid search.

Statistical evaluation methods

Continuous variables were first assessed for normality using the Shapiro-Wilk test. Variables meeting normal distribution criteria were described as mean with standard deviation, with inter-group comparisons using one-way ANOVA. Non-normally distributed variables were described as median (interquartile range), with inter-group comparisons using Kruskal-Wallis test. Categorical variables were described as frequency (percentage), with inter-group comparisons using chi-square test. When theoretical frequencies < 5 cells exceeded 20% or the total sample size was < 40, Fisher's exact test was used. All statistical analyses were performed using Python3.2, with $p < 0.05$ indicating statistically significant differences. Following model pre-triage, the system automatically generated a set of binary classification evaluation metrics, including Accuracy, Precision, Recall, F1-score, and Matthews Correlation Coefficient (MCC), to facilitate performance assessment. The confusion matrix was visualized as a heatmap to provide an intuitive comparison of the model's performance across different categories, generated using Python's matplotlib and seaborn libraries.

Ethics

This study was approved by the Ethics Committee of a triple-A hospital in China. Given its retrospective design and that no identifiable patient information was transmitted during LLMs testing, the Ethics Committee waived the requirement for informed consent. The study adheres to the principles of Declaration of Helsinki [26].

Results

Distribution shift in production deployment

Analysis of the patient characteristics (Table 1) identified a significant shift in emergency severity distribution towards higher-acuity cases in the online diagnostic data. This population demonstrated an elevated proportion of Level II patients (11.5% vs. 8.6% in training set and 6.6% in the test set) and a corresponding reduction in Level IV (Less-urgent) cases. This pattern suggests a phenomenon of data distribution shift, wherein the model encounters a more severe and clinically complex case-mix during production deployment than was represented in the development datasets. This finding underscores the importance of evaluating model robustness against such real-world shifts in patient populations.

Feature importance analysis

Quantitative analysis based on SHAP (Shapley Additive exPlanations) value ranking (Fig. 3) revealed that method of arrival (0.0917) and basis for judgement (0.0654) were the core features driving the model's triage decision-making, followed by vital sign measurements—pulse (0.0499), SBP (0.0310), and SpO_2 (0.0265). Age contributed moderately (0.0180), whereas body temperature and sex, with SHAP values of 0.0021 and 0.0031 (ranking 13th and 12th, respectively), had negligible influence. Triage-level specific analysis (Fig. 4) further revealed that Level I were predominantly driven by SpO_2 (0.0081), Level II relied on method of arrival (0.1700), while Level III/IV consistently prioritized basis for judgement (0.1077/0.1479) and pulse (0.0917/0.0983) as the primary predictive factors. Notably, body temperature and sex were secondary features across all levels. The SHAP analysis confirmed that the model's decisions relied primarily on objective severity indicators rather than demographic parameters, aligning with the clinical principle of prioritizing vital signs assessment in emergency medicine.

Model accuracy evaluation

The emergency triage model was trained three times independently, designated as V1, V2, and V3. The classification performance of each version was evaluated on both training and test sets using standard metrics: accuracy, precision, recall, and the F1-score. As summarized in Table 2, all versions yielded virtually identical performance on the test set, with key metrics such as accuracy consistently at 98.0%. The observed minor variations (e.g., F1-Score range: 95.5%–95.8%) are entirely contained within the widely overlapping 95% confidence intervals. Therefore, it is no evidence of a meaningful performance difference between the different versions of the emergency triage model.

In clinical emergency triage, recall is critical for minimizing the risk of missing critically ill patients, while

Table 1 The distribution of demographic characteristics

Variable	Training set *N* = 91,743	Test set *N* = 11,559	Online diagnostic set *N* = 10,000	*p*
Age (year), median (IQR)	46 (31-67)	53 (33-70)	53 (33-70)	<0.001
Female sex	51,644 (56.3)	7131 (61.7)	5515 (55.2)	0.038
Mode of arrival				
120 Ambulance	12,357 (13.4)	1079 (9.3)	1459 (14.6)	<0.001
Vital sign measurements				
Temperature (℃), median (IQR)	37.1 (36.5-38)	37.2 (36.6-38.1)	37.2 (36.7-38.1)	<0.001
Pulse rate (bpm), median (IQR)	87 (77-98)	87 (76-99)	88 (76-100)	<0.001
Systolic blood pressure (mmHg), standard deviation (SD)	138 (25.8)	135 (24.0)	137 (24.9)	<0.001
Diastolic blood pressure (mmHg), standard deviation (SD)	81 (15.3)	80 (17.40)	81(18.8)	<0.001
Respiratory rate (per min), median (IQR)	17 (15-18)	16 (13-18)	16 (13-18)	<0.001
Oxygen saturation (%), median (IQR)	98 (97-99)	98 (97-99)	98 (97-99)	<0.001
NRS pain score, median (IQR)	0 (0-1)	0 (0-2)	0 (0-2)	<0.001
Common chief complaints (Top 10)				
General complaints (e.g., fever)	20,812 (22.7)	3111 (26.9)	1626 (16.2)	<0.001
Gastrointestinal-related complaints	14,359 (15.7)	4487 (38.8)	3603 (36.0)	<0.001
Neurological-related complaints	12,224 (13.3)	162 (1.4)	476 (4.8)	<0.001
Respiratory-related complaints	9764 (10.6)	243(2.1)	690 (6.9)	<0.001
Injuries	9391 (10.2)	196(1.7)	641 (6.4)	<0.001
Cardiovascular-related complaints	8535 (9.3)	1684 (14.6)	2094 (20.9)	<0.001
Obstetrics and gynecology-related complaints	6258 (6.8)	525 (4.5)	339 (3.4)	<0.001
Otorhinolaryngological system-related complaints	4180 (4.7)	78 (0.7)	286 (2.9)	<0.001
High risk injury mechanism (e.g., fall)	292 (0.3)	2 (0)	0	<0.001
Urological-related complaints	1812 (1.9)	1032 (8.9)	154 (1.5)	<0.001
Emergency Severity Index				
Level I: Critical	684 (0.8)	83 (0.7)	85 (0.9)	<0.001
Level II: Severe	7920 (8.6)	766 (6.6)	1152 (11.5)	<0.001
Level III: Urgent	34,085 (37.2)	4364 (37.8)	4701 (47.0)	<0.001
Level IV: Less-Urgent	49,054 (53.4)	6346 (54.9)	4062 (40.6)	<0.001

Data are presented as number (percentage) of visits unless otherwise indicated

Data are presented as n (%), median (IQR), or mean (SD) as appropriate

Continuous data that were not normally distributed, such as pain scores, are presented as median (interquartile range)

precision is essential for the efficient allocation of limited medical resources. Comparative analysis revealed that V3 attained enhanced precision while maintaining a recall rate comparable to the highest-performing version (V1). Therefore, based on clinical safety and resource utilization efficiency, V3 (designated as Pre-Triage V3) is recommended as the preferred model for implementation in emergency triage systems. The total training time for this version was approximately 3 h and 4 min.

Comparison with baseline models

To comprehensively evaluate model performance, Pre-Triage V3 was compared with three classical baseline models. As shown in Table 3, all baseline models exhibited varying degrees of overfitting, manifested by substantial performance degradation from training to test sets. Notably, Random Forest displayed the most severe overfitting, with training accuracy reaching 99.4%, while test accuracy and recall dropped to 81.5% and 62.5%, respectively. By contrast, Pre-Triage V3 demonstrated minimal discrepancies between training and test set metrics, indicating stable generalization.

As shown in Table 4, Pre-Triage V3 achieved the highest F1-scores across all triage levels (I–IV). The baseline models exhibited severe performance degradation at Level II. For instance, Random Forest had a recall of only 11.4% (95% CI: 9.3%–13.8%), which is clinically unacceptable. At Level I, although Random Forest attained a precision of 96.7%, its recall of 69.9% implied risk of missing critically ill patients. In contrast, Pre-Triage V3 achieved an optimal balance between recall (90.0%) and precision (95.7%) at Level I, and reached an F1-score of 98.9% at Level IV where sample size was sufficient. It should be noted that the confidence intervals for Level I were substantially wider than those of other levels due to the limited sample size ($n = 83$), indicating the need for external validation with larger cohorts.

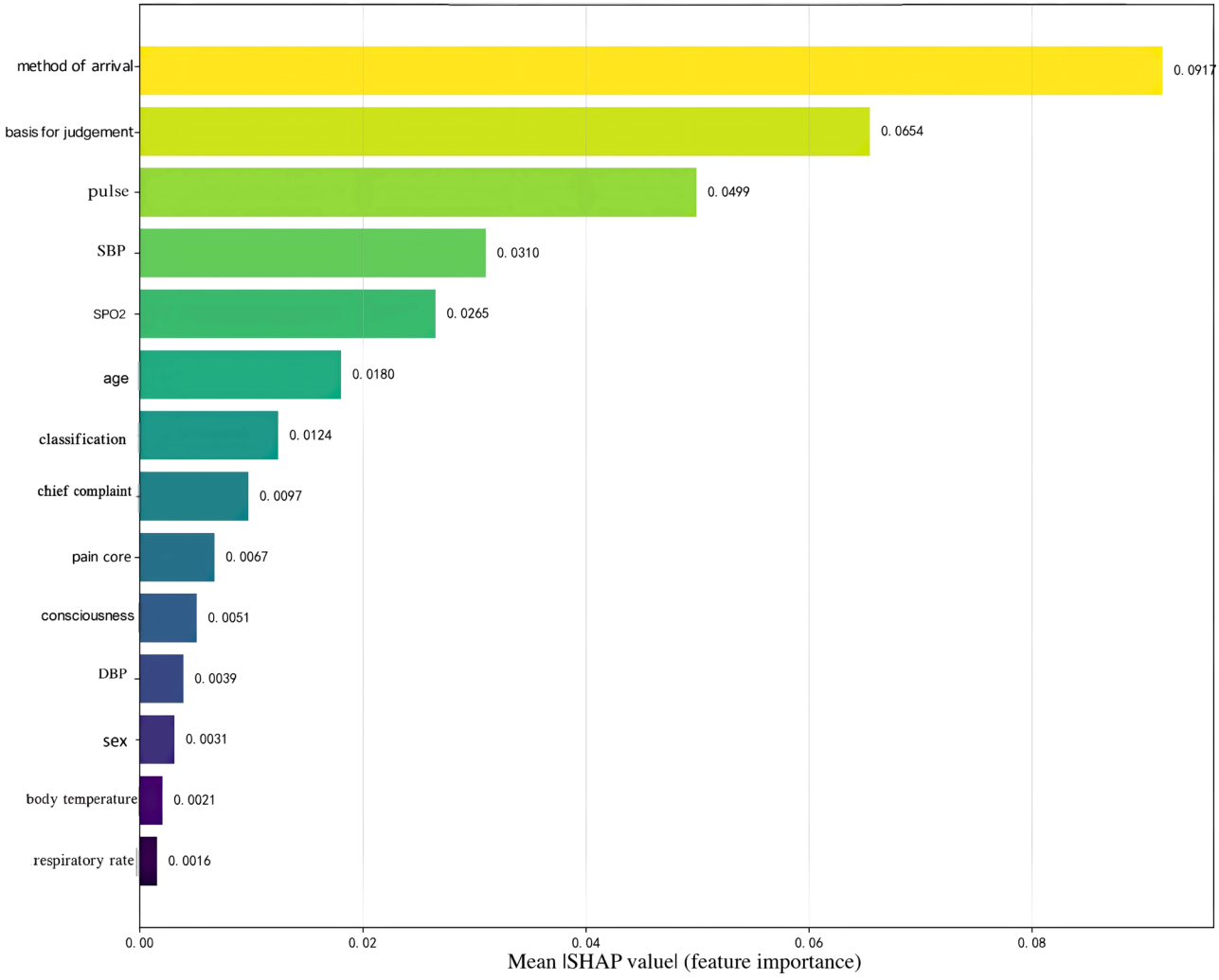

Fig. 3 Global feature mportance based on mean |SHAP| values

Results of online diagnostic test

The model was evaluated using the online diagnostic data ($N = 10{,}000$), which included 85 cases of Level I, 1,152 cases of Level II, 4,701 cases of Level III and 4,062 cases of Level IV. The results demonstrated that the Pre-Triage V3 achieved an overall accuracy of 89.45%. The data was partitioned into 10 equal subsets to generate a trend chart of model accuracy (Fig. 5).

Detailed performance across the four triage levels is presented in Table 5, where the model achieved classification accuracies ranging from 90.0% to 99.6%. The MCC reached 0.92, indicating adequate performance and strong agreement between predicted and actual classifications. A confusion matrix heatmap illustrating the model's performance across all categories is shown in Fig. 6.

Online diagnostic test demonstrated a 76.5% recall for Level I (critical) cases, indicating that nearly one-quarter of critical cases were not prioritized by the system. Given the severe consequences of missed critical diagnoses, we conducted a threshold sensitivity analysis (Fig. 7). As shown, adjusting the classification threshold from 0.50 to 0.10 increased Level I recall from 88.2% to 97.6% (meeting the safety criterion) and reduced missed diagnoses by 80%. Clinical risk assessment under a cost matrix assigning 20-fold greater weight to missed diagnosis versus over-triage revealed a 60% reduction in overall risk score, substantiating the algorithmic viability of prioritizing high recall for critical cases.

External validation

External validation was performed on 4,504 triage records from another tertiary hospital in the same region (December 11–18, 2025). The model achieved an accuracy of 78.3% (95% CI: 76.9-79.8%), representing a 11.2% performance degradation compared to the online diagnostic set.

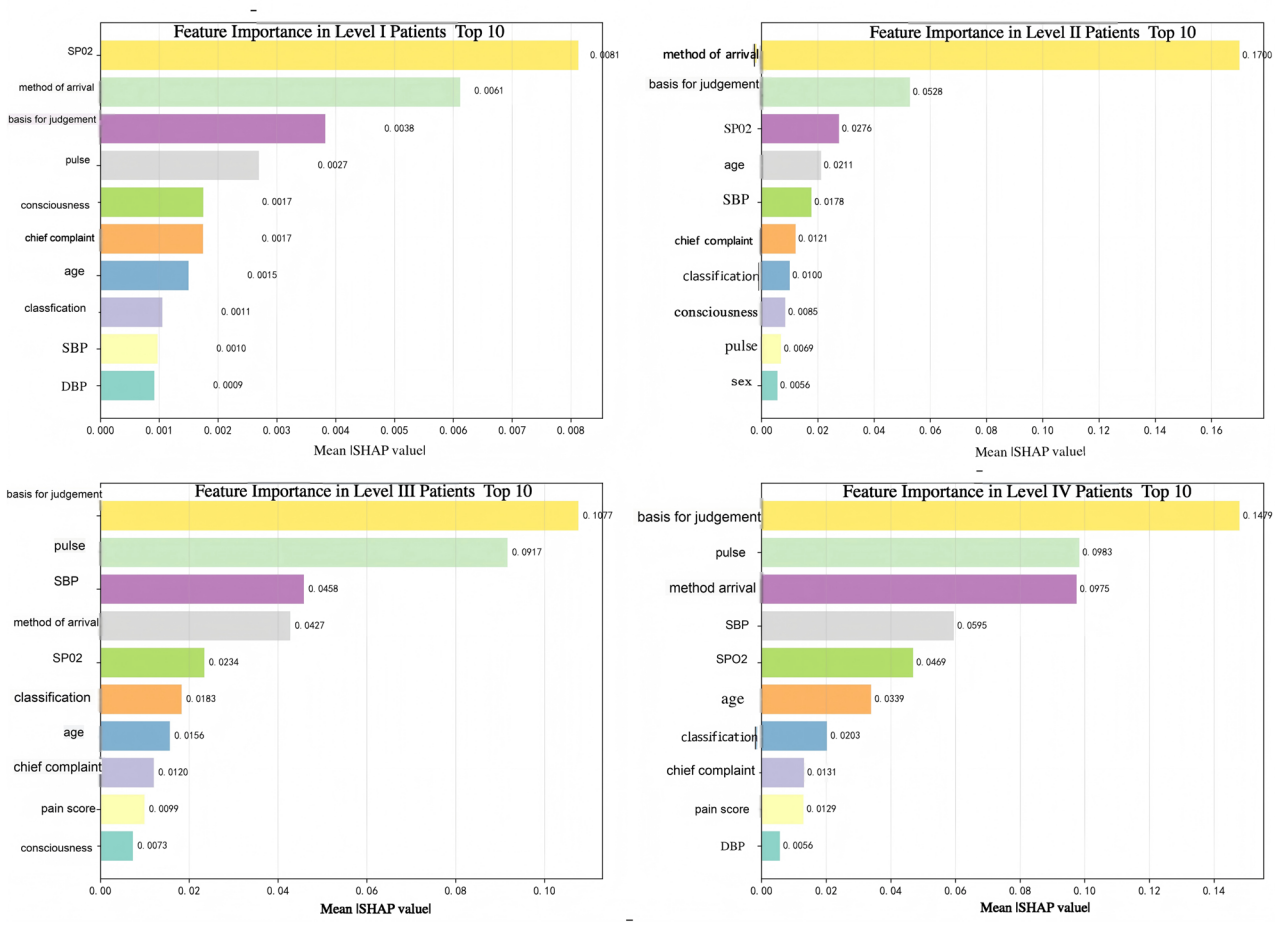

Fig. 4 Comparative analysis of feature importance across triage levels

Table 2 Performance comparison of model versions 1.0 to 3.0

Version	Type	Accuracy(95% CI)	Precision(95% CI)	Recall(95% CI)	F1- score(95% CI)
1.0	Training Set	97.2%(96.6%-97.5%)	95.5%(95.0%-96.0%)	93.6%(93.0%-94.2%)	94.5%(94.0%-95.0%)
	Test Set	98.0%(97.7%-98.3%)	95.6%(95.0%-96.2%)	95.6%(95.0%-96.2%)	95.6%(95.1%-96.1%)
2.0	Training Set	97.3%(97.0%-97.6%)	96.2%(95.7%-96.7%)	92.8%(92.2%-93.4%)	94.4%(93.9%-94.9%)
	Test Set	98.0%(97.7%-98.3%)	96.5%(96.0%-97.0%)	94.4%(93.8%-95.0%)	95.5%(95.0%-96.0%)
3.0	Training Set	97.3% (97.0%-97.6%)	96.0%(95.4%-96.6%)	93.2% (92.6%-93.8%)	94.5%(94.0%-95.0%)
	Test Set	98.0% (97.4%-98.6%)	96.1%(95.6%-96.6%)	95.5% (94.9%-96.1%)	95.8%(95.3%-96.3%)

Table 3 Performance comparison of baseline model

Version	Type	Accuracy (95%CI)	Precision (95%CI)	Recall (95% CI)	F1-score (95% CI)
Logistic Regression	Training Set	91.0% (90.8%-91.2%)	77.5% (76.9%-78.1%)	86.5% (85.8%-87.2%)	79.5% (78.8%-80.2%)
	Test Set	86.9% (86.2%-87.6%)	71.8% (70.6%-73.1%)	79.2% (76.4%-81.5%)	70.6% (68.9%-72.2%)
Random Forest	Training Set	99.4% (99.3%-99.4%)	98.5% (98.1%-98.8%)	99.6% (99.5%-99.6%)	99.0% (98.8%-99.2%)
	Test Set	81.5% (80.8%-82.28%)	87.3% (85.2%-89.1%)	62.5% (59.9%-65.1%)	66.7% (64.5%-68.7%)
XGBoost	Training Set	94.6% (94.4%-94.7%)	92.3% (91.9%-92.8%)	93.9% (93.7%-94.1%)	93.1% (92.8%-93.4%)
	Test Set	83.7% (83.1%-84.4%)	78.2% (76.3%-79.8%)	84.2% (82.1%-86.2%)	78.1% (76.7%-79.6%)

Discussion

Interpretation of data distribution shift

The four-level triage protocol remained stable throughout the study (see Methods). Therefore, the observed distributional shifts in Table 1 can be attributed to genuine evolution in the patient population—such as seasonal fluctuations, demographic changes, and variations in case-mix—rather than bias from alterations in triage

Table 4 Performance of baseline model across four emergency triage levels

Triage	Version	Precision (95% CI)	Recall (95% CI)	F1-score (95% CI)
Level I: Critical (83 cases)	Logistic Regression	19.2% (12.1%-28.9%)	77.1% (67.0%-84.8%)	30.7% (21.8%-41.3%)
	Random Forest	96.7% (90.3%-98.9%)	69.9% (59.3%-78.7%)	81.1% (73.5%-88.7%)
	XGBoost	91.1% (83.1%-95.6%)	86.8% (77.8%-92.4%)	88.9% (80.3%-94.0%)
	Pre-TriageV3	95.7% (89.3%-98.4%)	90.0% (81.7%-94.8%)	92.8% (86.2%-96.4%)
Level II: Severe (766 cases)	Logistic Regression	89.7% (87.3%-91.6%)	65.7% (62.2%-68.9%)	75.8% (72.7%-78.7%)
	Random Forest	87.9% (85.4%-90.0%)	11.4% (9.3%-13.8%)	20.1% (17.4%-23.1%)
	XGBoost	31.1% (27.9%-34.4%)	82.5% (79.7%-85.0%)	45.1% (41.7%-48.7%)
	Pre-TrigeV3	92.3% (90.3%-94.0%)	86.7% (84.2%-88.9%)	89.4% (87.2%-91.4%)
Level III: Urgent (4364 cases)	Logistic Regression	90.4% (89.5%-91.3%)	78.9% (77.6%-80.1%)	84.3% (83.1%-85.3%)
	Random Forest	85.0% (83.9%-86.0%)	72.6% (71.3%-73.9%)	78.3% (77.1%-79.5%)
	XGBoost	97.1% (96.6%-97.6%)	83.6% (82.4%-84.6%)	89.8% (88.9%-90.7%)
	Pre-TriageV3	96.5% (95.9%-97.1%)	97.0% (96.5%-97.5%)	96.8% (96.3%-97.3%)
Level IV: Non-Urgent (6346 cases)	Logistic Regression	88.1% (87.2%-88.8%)	95.2% (94.6%-95.7%)	91.5% (90.8%-92.1%)
	Random Forest	79.6% (78.6%-80.6%)	96.2% (95.7%-96.6%)	87.1% (86.3%-87.9%)
	XGBoost	93.6% (92.9%-94.1%)	83.9% (83.0%-84.8%)	88.5% (87.6%-89.2%)
	Pre-TriageV3	98.5% (98.2%-98.8%)	99.2% (99.0%-99.4%)	98.9% (98.6%-99.1%)

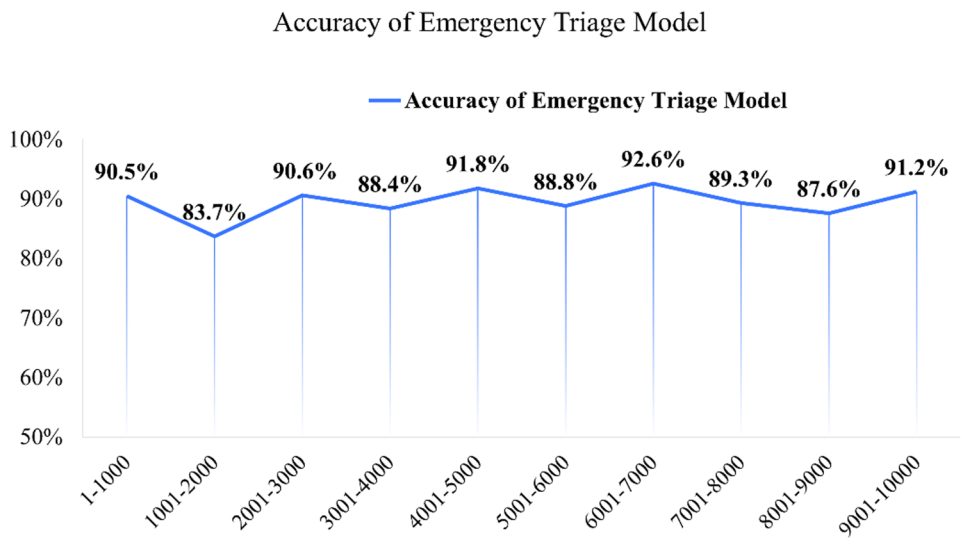

Fig. 5 Accuracy trend of Pre-Triage V3

Table 5 Performance of Pre-Triage V3 in online diagnostic test

Triage	Accuracy(95%CI)	Precision(95%CI)	Recall(95%CI)%	F1-score(95%CI)
Level I: Critical	99.6%(99.4%-99.8%)	75.6%(70.2%-80.9%)	76.5%(71.1%-81.8%)	76.0%(71.0%-81.0%)
Level II: Severe	97.9%(97.5-98.3%)	92.1%(90.5-93.6%)	89.8%(88.0-91.5%)	90.9%(89.5-92.3%)
Level III: Urgent	90.0%(89.2-90.9%)	95.8%(95.0-96.5%)	82.4%(81.2-83.7%)	88.6%(87.6-89.6%)
Level IV: Less-Urgent	91.4%(90.5-92.2%)	83.7%(82.3-85.1%)	97.8%(97.2-98.3%)	90.2%(89.2-91.1%)

rules. This reflects the dynamic nature of clinical practice. The model's stable performance across this evolving population demonstrates its robustness and generalization capacity in the face of data distribution changes, with no evidence of overfitting to the static training set.

Model performance and clinical implications

In this study, a dedicated emergency triage model was developed using structured text data extracted from electronic medical records, and implemented based on the ERNIE LLM. Compared with traditional statistical methods and standard big data approaches, LLMs provide the advantage of multidimensional analysis and rapid information processing. These models can assist in extracting disease characteristics and clinical manifestations, support decision-making processes, and potentially reduce subjective judgement biases [27]. Three independently trained versions of the triage

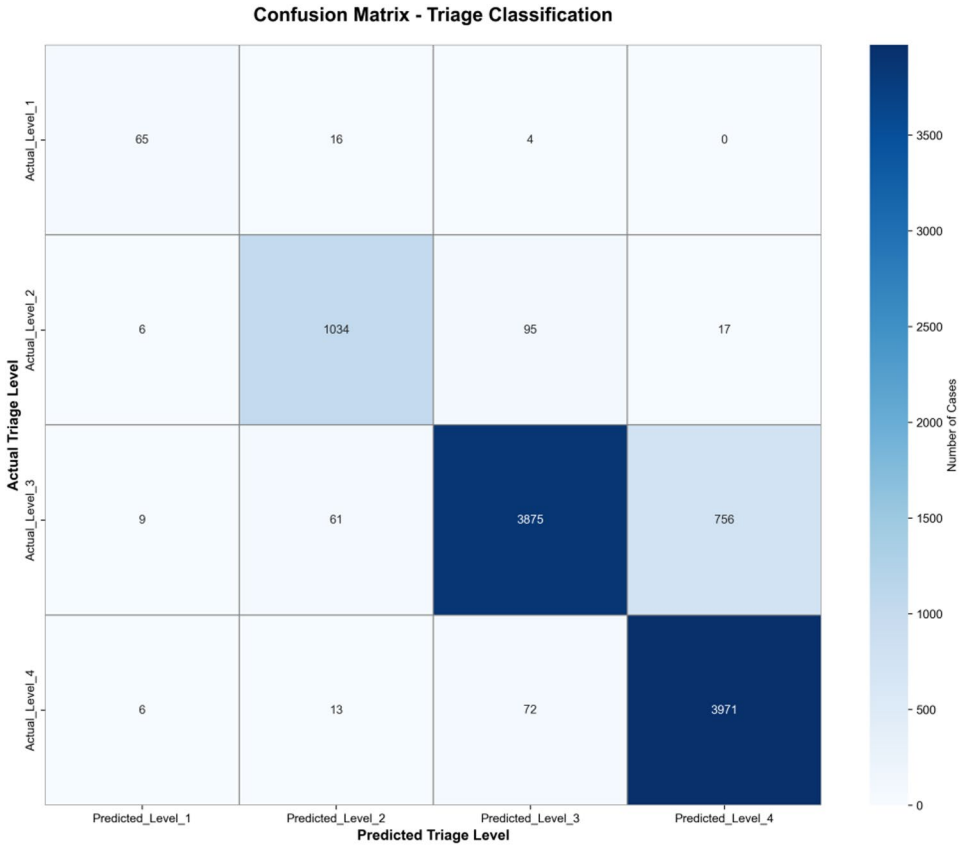

Fig. 6 Confusion matrix heatmap of Pre-Triage V3

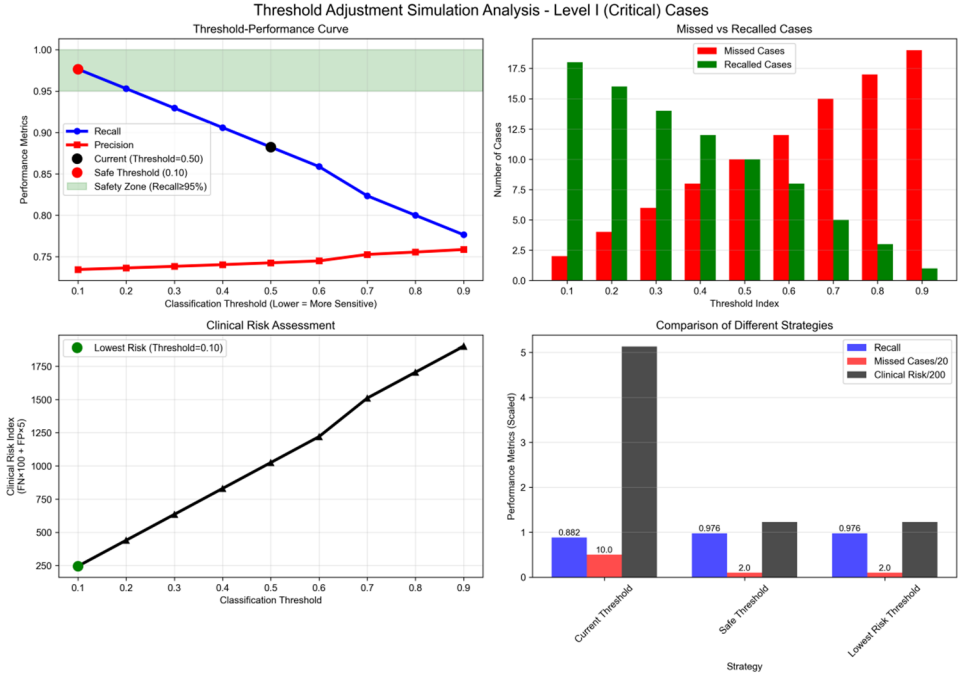

Fig. 7 Threshold sensitivity analysis

model demonstrated highly consistent performance, with test set accuracy reaching 98.0% (95% CI: 97.7%-98.3%) across all versions. The minimal performance variation and overlapping confidence intervals indicate training stability and reproducibility. While these results demonstrate the model's accuracy and reliability on real-world patient data, it is important to note that our data cleaning process, although consistent in formatting and handling of missing values, did not involve deduplication—a deliberate decision aimed at preserving clinically distinct patient visits. The model's performance may thus be attributed to the reduction of noise through standardized preprocessing procedures, even as we acknowledge the inherent limitations discussed in the following sections. For contextual comparison, Zhen et al. utilized a machine learning model (extreme gradient boosting algorithm) trained on historical information system data, achieving a triage accuracy of 82.57% [28]. Notably, the F1 scores for Level III and Level IV triage classifications exceeded 96% across all model versions, demonstrating strong performance in identifying patients with lower acuity.

Comparison of accuracy across different triage systems

Following deployment, the proposed model achieved an overall triage accuracy of 89.45%, which is notably higher than the 76% accuracy reported for ChatGPT-4.0 in prior literature [29]. This favorable performance underscores the enhanced adaptability of localized models trained on targeted datasets. While these results are promising, it is crucial to position this technology strictly as an assistive tool within clinical workflows. The models exhibited lower Recall (86.7%) and F1-scores (89.4%) for Level II cases, likely attributable to symptom overlap with adjacent levels (e.g., chest pain, dyspnea) despite differing clinical severity, which obscured decision boundaries during training. This pattern mirrors Christopher's study [14], which achieved 100% accuracy for distinct categories (Less-Urgent vs. Critical) but only 86% and 75% for Level I-II and II-III differentiation, respectively, confirming that model performance degrades as triage levels become less distinguishable. To address this limitation, it is suggested to adopt the "Human-computer cooperation" mode, in which AI completes initial screening and nurses subsequently review key cases such as Level II acute and severe instances. This complementary approach optimizes overall triage efficiency and accuracy.

Post-deployment performance evaluation

The tool holds potential for improving workforce allocation within emergency departments and streamlining clinical workflows. After deployment, the model enables even professional users to assess the urgency of illness based on existing symptoms and clinical history, thereby guiding appropriate care-seeking decisions [30]. Although rule-based digital triage systems [31] are currently the most widely used, their core relies on a predefined, static logical framework. When encountering dynamic disease patterns or atypical cases with complex symptom combinations, such systems struggle to adapt, resulting in limited flexibility and adaptability. General-purpose LLMs lack transparency regarding training data and internal architectures, leading to considerable variability in diagnostic performance across different clinical tasks [16, 32, 33]. Given its high performance and ease of implementation, domain-specific, self-trained models continue to represent a promising and scalable direction in the development of diagnostic tools for emergency triage.

Sensitivity analysis revealed that threshold adjustment converts the model to a high-sensitivity screening instrument, operationalizing the emergency triage principle of minimizing missed diagnoses at the cost of increased false positives. While this elevates nursing workload through over-triage, it substantially reduces the risk of undetected critical cases.

The observed performance drop in external validation likely stems from inter-institutional variability in chief complaint documentation and triage protocol implementation. Qualitative analysis revealed significant heterogeneity in free-text chief complaint descriptions and subtle differences in severity thresholds between sites, highlighting the model's sensitivity to data distribution shifts. Therefore, it is important to focus on both standardized mapping relationships that align free-text chief complaints with canonical triage categories and multi-center protocol harmonization to enhance cross-institutional generalizability.

SHAP value analysis elucidated the internal decision-making rationale of the triage model. The five most influential features were method of arrival, basis for judgement, pulse, SBP, and SPO_2 (Fig. 3). This finding yields two key clinical insights. Firstly, while the prominence of vital signs (pulse, SBP, SPO_2) confirms the model's alignment with the "vital signs first" principle, a cornerstone emphasized in both the Expert Consensus on Emergency Triage [22] and the TTAS [23] standard. The high importance of non-physiological features (method of arrival, judgement basis) reveals a hybrid decision logic that also incorporates contextual and operational data. This nuanced alignment may enhance clinicians' trust in the AI system. Second, the quantitative feature ranking provides empirical grounds for workflow optimization. For instance, the strong influence of "method of arrival" suggests that triage protocols should ensure its consistent and accurate documentation, as it serves as a powerful proxy for pre-hospital acuity. Conversely, the relatively lower importance of "chief complaint" (rank: 8th) indicates that the current model,

which relies primarily on structured data, may underutilize the rich semantic information in free-text patient descriptions.

Although SHAP cannot inherently distinguish causality, future research should incorporate the Target Trial Emulation framework, encoding clinical causal knowledge into a Directed Acyclic Graph (DAG) to constrain model structure. This would guide SHAP to focus on stable causal features (e.g., consciousness level, SpO_2) rather than confounding signals, thereby substantially enhancing clinical interpretability and intervention robustness while maintaining predictive performance.

Limitations

Although the proposed model outperforms the baseline models (Logistic Regression, Random Forest, and XGBoost), it should be noted that its structural complexity and the additional computational overhead from feature extraction prolong training time and maintain a high dependence on manual feature engineering, greatly increasing the operational cost. In the future, we plan to extend the current statistical validation framework by rigorously evaluating the model's generalizability across multi-center datasets and its robustness to domain shifts in real-world clinical settings. The above work will effectively improve the scientific rigor and reliability of the results of model evaluation, and provide strong support for its promotion and application in clinical practice.

Conclusion

This study successfully constructed and validated an emergency pre-triage model based on the EasyDL all-in-one development platform. The model demonstrated adequate performance and can be deployed to mobile terminals for clinical application. The development process proved operationally feasible, and the resulting model holds significant practical value for clinical use, drawing greater attention from healthcare professionals to the immense potential of AI technologies and tools in traditional medical practice.

Future work

We acknowledge that the current model's performance in classifying severe cases (Levels I & II) remains suboptimal. This limitation is likely attributable to the constrained representational capacity of the existing feature set in capturing complex, non-linear, and atypical symptom presentations. To address this critical shortfall, our subsequent research will focus on integrating richer, multimodal data sources—particularly unstructured, free-text symptom descriptions from triage notes. By leveraging advanced natural language processing (NLP) techniques for nuanced feature extraction, we aim to enhance the model's semantic understanding, improve its generalizability across a wider spectrum of clinical presentations, and ultimately increase its clinical relevance and utility for high-acuity patient care. Although the model demonstrates consistent performance across seasons, temporal biases may still arise when data are segmented by time period. Future studies should incorporate seasonal covariates and conduct cross-year validation analyses to quantify and adjust for such cyclical variations, thereby enhancing the model's generalizability and long-term clinical applicability.

Abbreviations

API	Application Programming Interface
MCC	Matthews Correlation Coefficient
LLMs	Large Language Models
GPT	Generative Pre-trained transformer
RAG	Retrieval Augmented Generation
AI	Artificial Intelligence
ID	Identification
TTAS	The Taiwan Triage and Acuity Scale
SPO2	Saturation of peripheral oxygen
SBP	Systolic blood pressure
DBP	Diastolic blood pressure
ERNIE	Enhanced Representation through kNowledge intEgration
GPU	Graphics Processing Unit
GB	Gigabyte
VRAM	Video Random Access Memory
CPU	Central Processing unit
RAM	Random Access Memory
FLOPS	Floating-Point Operations Per Second
XGBoost	Extreme Gradient Boosting
NLP	Natural Language Processing

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12873-026-01489-9.

Supplementary Material 1

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Not applicable.

Author contributions

YJG and XNC proposed the main research objectives, were responsible for the study conception and design, research implementation, and manuscript writing. GXS and YRG performed data collection and organization, statistical analysis, as well as figure and table preparation. JT and YX were responsible for participant selection, sample collection, and data annotation. TQS and PH oversaw quality control and review of the manuscript, took overall responsibility for the article, and supervised the research process. All authors read and approved the final manuscript.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
This study is a retrospective analysis and all participants were exempted from informed consent. The research was approved by the Ethics Committee of Nanjing Drum Tower Hospital, Affiliated Hospital of Medical School, Nanjing University (Approval Number: 2025-0302-01).

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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References
1. Çetin SB, Cebeci F, Eray O. The effect of computer-based decision support system on emergency department triage: Non-randomised controlled trial. Int Emerg Nurs. 2023;70:101341doi. https://doi.org/10.1016/j.ienj.2023.101341.
2. Zhang LY, Lang HJ, Xu L, et al. Research advances in emergency Pre - triage system and information management. J Nurs Sci. 2020;34(24):4414–9. https://doi.org/10.12102/j.issn.1009-6493.2020.24.020.
3. Zhou X, Zheng RF, Jian GR, et al. Advancements in emergency triage for patients presenting with chest pain. Chin J Nurs. 2022;57(16):2036–41. https://doi.org/10.3761/j.issn.0254-1769.2022.16.018.
4. Ge J, Lai JC. Artificial intelligence-based text generators in hepatology: ChatGPT is just the beginning. Hepatol Commun. 2023;7(4):e0097. https://doi.org/10.1097/HC9.0000000000000097.
5. Ji Z, Lee N, Frieske R, et al. Survey of hallucination in natural Language generation. ACM-CSUR. 2023;55(12):248. https://doi.org/10.1145/3571730.
6. Lyons RJ, Arepalli SR, Fromal O, Choi JD, Jain N. Artificial intelligence chatbot performance in triage of ophthalmic conditions. Canadian journal of ophthalmology. J Canadien D'ophtalmologie. 2024;59(4):e301–8. https://doi.org/10.1016/j.jcjo.2023.07.016.
7. Delsoz M, Madadi Y, Raja H, et al. Performance of ChatGPT in diagnosis of corneal eye diseases. Cornea. 2024;43(5):664–70. https://doi.org/10.1097/ICO.0000000000003492.
8. Liu X, Wu J, Shao A et al. Transforming retinal vascular disease classification: a comprehensive analysis of chatgpt's performance and inference abilities on non-English clinical environment. medRxiv2023;:2023-06.doi: https://doi.org/10.1101/2023.06.28.23291931
9. Hirosawa T, Harada Y, Yokose M, et al. Diagnostic accuracy of Differential-Diagnosis lists generated by generative pretrained transformer 3 chatbot for clinical vignettes with common chief complaints: A pilot study. Int J Environ Res Public Health. 2023;20(4):3378. https://doi.org/10.3390/ijerph20043378.
10. Hirosawa T, Kawamura R, Harada Y, et al. ChatGPT-Generated differential diagnosis lists for complex Case-Derived clinical vignettes: diagnostic accuracy evaluation. JMIR Med Inf. 2023;11:e48808. https://doi.org/10.2196/48808.
11. Nakaura T, Yoshida N, Kobayashi N, et al. Preliminary assessment of automated radiology report generation with generative pre-trained transformers: comparing results to radiologist-generated reports. Japanese J Radiol. 2024;42(2):190–200. https://doi.org/10.1007/s11604-023-01487-y.
12. Ito N, Kadomatsu S, Fujisawa M, et al. The accuracy and potential Racial and ethnic biases of GPT-4 in the diagnosis and triage of health conditions: evaluation study. JMIR Med Educ. 2023;9:e47532. https://doi.org/10.2196/47532.
13. Paslı S, Şahin AS, Beşer MF, et al. Assessing the precision of artificial intelligence in ED triage decisions: insights from a study with ChatGPT. Am J Emerg Med. 2024;78:170–5. https://doi.org/10.1016/j.ajem.2024.01.037.
14. Williams CYK, Zack T, Miao BY, et al. Use of a large Language model to assess clinical acuity of adults in the emergency department. JAMA Netw Open. 2024;7(5):e248895. https://doi.org/10.1001/jamanetworkopen.2024.8895.
15. Yazaki M, Maki S, Furuya T, et al. Emergency patient triage improvement through a Retrieval-Augmented generation enhanced Large-Scale Language model. Prehospital Emerg Care. 2025;29(3):203–9. https://doi.org/10.1080/10903127.2024.2374400.
16. Statz GM, Evans AZ, Johnston SL, et al. Can artificial intelligence enhance syncope management? A JACC: advances multidisciplinary collaborative statement. JACC Adv. 2023;2(3):100323. https://doi.org/10.1016/j.jacadv.2023.100323.
17. Fellegi IP, Sunter AB. A theory for record linkage. J Am Stat Assoc. 1969;64(328):1183–210. https://doi.org/10.1080/01621459.1969.10501049.
18. Mudgal S, Li H, Rekatsinas T, et al. Deep learning for entity matching: a design space exploration. In Proceedings of the 2018 international conference on management of data. 2018;19–34).https://doi.org/10.1145/3183713.3196926
19. Davenport TH, Prusak L. Working knowledge: how organizations manage what they know. Harvard Business Review; 1998.
20. Li Y, Li J, Suhara Y, et al. Deep Entity Matching with Pre-Trained Language Models. Proceedings of the VLDB Endowment. 2020;14(1):50–60.
21. Chen QJ, Fang F, Huang P, et al. Design and application of an emergency triage information system. Chin J Nurs. 2015;50(8):963–6. https://doi.org/10.3761/j.issn.0254-1769.2015.08.015.
22. Expert Consensus Group on Emergency Pre-triage. Expert consensus on emergency triage: A preliminary review. Chin J Emerg Med. 2018;27(6):599–604. https://doi.org/10.3760/cma.j.issn.1671-0282.2018.06.006.
23. Chiu HY, Chen LC, Lin XZ, et al. Current trends in emergency triage in taiwan: the Five-Level triage system. J Nurs Sci. 2008;55(3):87–91. https://doi.org/10.6224/JN.55.3.87.
24. Ministry of Health of the People's Republic of China. Guiding principles for grading the condition of emergency patients (draft for soliciting opinions). Chin J Crit Care Med (Electronic Edition). 2011;4(4):241–3. https://doi.org/10.3877/cma.j.issn.1674-6880.2011.04.006.
25. Chen QJ, Fang F, Huang P, et al. Construction and application of emergency triage information system based on TTAS standard. Chin Nurs Res. 2015;29(32):4050–2. https://doi.org/10.3969/j.issn.1009-6493.2015.32.029.
26. World Medical Association Declaration of Helsinki: ethical principles for medical research involving human subjects. JAMA. 2013;310(20):2191–4. https://doi.org/10.1001/jama.2013.281053.
27. Preiksaitis C, Ashenburg N, Bunney G, et al. The Role of large language models in transforming emergency medicine: Scoping Review. JMIR medical informatics. 2024;12:e53787. https://doi.org/10.2196/53787
28. Gao Z, Qi X, Zhang X, et al. Developing and validating an emergency triage model using machine learning algorithms with medical big data. Risk Manage Healthc Policy. 2022;15:1545–51. https://doi.org/10.2147/RMHP.S355176.
29. Fraser H, Crossland D, Bacher I, et al. Comparison of diagnostic and triage accuracy of Ada health and WebMD symptom Checkers, ChatGPT, and physicians for patients in an emergency department: clinical data analysis study. JMIR mHealth and. 2023;11:e49995. https://doi.org/10.2196/49995.
30. Chenais G, Lagarde E, Gil-Jardiné C. Artificial intelligence in emergency medicine: viewpoint of current applications and foreseeable opportunities and challenges. J Med Internet Res. 2023;25:e40031. https://doi.org/10.2196/40031.
31. Gu QY, Li XY, Fan XY, Zhao J. Study on optimizing the flowsheet visiting doctors' office and triage standards in emergency department. Chin Nurs Res. 2010;24(34):3163–5. https://doi.org/10.3969/j.issn.1009-6493.2010.34.032.
32. OpenAI, Achiam J, Adler S, et al. GPT – 4 technical report. arXiv. Preprint posted online Mar 27. 2023. https://doi.org/10.48550/arXiv.2303.08774
33. Brin D, Sorin V, Vaid A, et al. Comparing ChatGPT and GPT-4 performance in USMLE soft skill assessments. Sci Rep. 2023;13(1):16492. https://doi.org/10.1038/s41598-023-43436-9.

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